

WE Prerequisite: Outdoor Water Use Reduction

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To reduce outdoor potable water consumption and preserve no and low-cost potable water resources.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Reduce outdoor water use through one of the following options. Nonvegetated surfaces, such as permeable or impermeable pavement, should be excluded from the landscape area calculations. Athletic fields and playgrounds (if vegetated) and food gardens may be included or excluded at the project team's discretion.

Option 1. No Irrigation Required

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

Option 2. Reduced Irrigation

Reduce the project's landscape water requirement by at least 30% from the calculated baseline for the site's peak watering month. Reductions must be achieved through plant species selection and irrigation system efficiency, as calculated by the Environmental Protection Agency (EPA) WaterSense Water Budget Tool.

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Designing out the need for a permanent irrigation system, selecting native or adapted plants for project landscaping and/or installing efficient irrigation systems reduces irrigation water use throughout the building life cycle, which may help improve a project's water performance score.

WE Prerequisite: Indoor Water Use Reduction

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To reduce indoor potable water consumption and preserve no and low cost potable water resources.

REQUIREMENTS

NC, CS, SCHOOLS, NC-RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, NC-HOSPITALITY, HEALTHCARE

Building Water Use

For the fixtures and fittings listed in Table 1, as applicable to the project scope, reduce aggregate water consumption by 20% from the baseline. Base calculations on the volumes and flow rates shown in Table 1.

All newly installed toilets, urinals, private lavatory faucets, and showerheads that are eligible for labeling must be WaterSense labeled (or a local equivalent for projects outside the U.S.).

Table 1. Baseline water consumption of fixtures and fittings

<i>Fixture or fitting</i>	<i>Baseline (IP units)</i>	<i>Baseline (SI units)</i>
Toilet (water closet)*	1.6 gpf	6 lpf
Urinal*	1.0 gpf	3.8 lpf
Public lavatory (restroom) faucet	0.5 gpm at 60 psi** all others except private applications	1.9 lpm at 415 kPa, all others except private applications
Private lavatory faucets	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Kitchen faucet (excluding faucets used exclusively for filling operations)	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Showerhead*	2.5 gpm at 80 psi per shower stall	9.5 lpm at 550 kPa per shower stall

* WaterSense label available for this product type
 gpf = gallons per flush
 gpm = gallons per minute
 psi = pounds per square inch

lpf = liters per flush
 lpm = liters per minute
 kPa = kilopascals

Projects located where standard supply pressure is different than the LEED baseline supply pressure may calculate the water consumption of flow fixtures and fittings at the local standard supply pressure.

Appliance and Process Water Use

Install appliances, equipment, and processes within the project scope that meet the requirements listed in the tables below.

Existing appliances intended for reuse in the project are not required to meet the requirements in Table 2.

Table 2. Standards for appliances

<i>Appliance</i>	<i>Requirement</i>
Residential clothes washers	ENERGY STAR or performance equivalent*
Commercial clothes washers	ENERGY STAR or performance equivalent
Residential dishwashers (standard and compact)	ENERGY STAR or performance equivalent*
Prerinse spray valves	≤ 1.3 gpm (4.9 lpm)
Ice machine	ENERGY STAR or performance equivalent and use either air-cooled or closed-loop cooling, such as chilled or condenser water system

gpm = gallons per minute

lpm = liters per minute

*Projects in Europe may install residential appliances meeting the EU A label.

Table 3. Standards for processes

<i>Process</i>	<i>Requirement</i>
Heat rejection and cooling	No once-through cooling with potable water for any equipment or appliances that reject heat
Cooling towers and evaporative condensers	Equip with • makeup water meters • conductivity controllers and overflow alarms • efficient drift eliminators that reduce drift to maximum of 0.002% of recirculated water volume for counterflow towers and 0.005% of recirculated water flow for cross-flow towers

Healthcare, Retail, Schools, and Hospitality Only

In addition, water-consuming appliances, equipment, and processes must meet the requirements listed in Tables 4 and 5.

Table 4. Standards for appliances

<i>Kitchen equipment</i>	<i>Requirement (IP units)</i>	<i>Requirement (SI units)</i>
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Dishwasher	Undercounter	≤ 1.6 gal/rack	≤ 6.0 liters/rack
	Stationary, single tank, door	≤ 1.4 gal/rack	≤ 5.3 liters/rack
	Single tank, conveyor	≤ 1.0 gal/rack	≤ 3.8 liters/rack
	Multiple tank, conveyor	≤ 0.9 gal/rack	≤ 3.4 liters/rack
	Flight machine	≤ 180 gal/hour	≤ 680 liters/hour
Food steamer	Batch	≤ 6 gal/hour/pan	≤ 23 liters/hour/pan
	Cook-to-order	≤ 10 gal/hour/pan	≤ 38 liters/hour/pan
Combination oven,	Countertop or stand	≤ 3.5 gal/hour/pan	≤ 13 liters/hour/pan
	Roll-in	≤ 3.5 gal/hour/pan	≤ 13 liters/hour/pan

Table 5. Process requirements

Discharge water temperature tempering	Where local requirements limit discharge temperature of fluids into drainage system, use tempering device that runs water only when equipment discharges hot water OR Provide thermal recovery heat exchanger that cools drained discharge water below code-required maximum discharge temperatures while simultaneously preheating inlet makeup water OR If fluid is steam condensate, return it to boiler
Venturi-type flow-through vacuum generators or aspirators	Use no device that generates vacuum by means of water flow through device into drain

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

WE prerequisite Indoor Water Use Reduction incorporates feedback from and market barriers identified by LEED project teams. The intent of these changes is to make prerequisite compliance more achievable and relevant to international project teams by recognizing regional variations while maintaining the stringency of requirements for water performance.

Further Explanation

Refer to the LEED v4 reference guide, with the following additions:

Occupant Types

If the project includes separate gender neutral and/or ADA restrooms without urinals, assume that 5% of male occupants and 5% of female occupants use these restrooms. Enter 95% into the percent of males expected to use restrooms with urinals in the Indoor Water Use Reduction Calculator. Alternately, estimate this percentage based on the project's restroom layout and anticipated usage patterns or weighted fixture counts.

International Tips

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ International projects located where standard water supply pressure is different than the LEED baseline supply pressure values may calculate the water consumption of flow fixtures and fittings at the local standard water supply pressure. Product cutsheets must demonstrate that the fixture or fitting complies with the LEED baseline flow rate.
 - Projects that are unable to provide manufacturer documentation of the fixture flow rate at the LEED baseline water supply pressure may use the local standard supply water pressure in the design case and the standard LEED baseline flow rates in the baseline case.
 - Projects served by water supply pressures different than specified in LEED may install pressure compensating aerators in flow fixtures to achieve the desired flow rate without compromising user satisfaction.
- ▶ Where local building code conflicts with LEED prescriptive requirements, project teams may select fixtures that comply with local code and compensate by selecting more water-efficient fixtures for other fixture types, provided that the flow rate for the fixture is the lowest allowable by code, and the project meets in aggregate the required 20% reduction from LEED baseline. Provide an excerpt of the local code highlighting the flow rate requirements when submitting for review.
- ▶ The EU A label for residential appliances is an acceptable alternative to ENERGY STAR. The EU Ecodesign and Labelling framework establishes minimum performance standards for the energy and environmental performance of appliances and products.
- ▶ The CEE Commercial Clothes Washer Specification is no longer active. Commercial clothes washers require the ENERGY STAR label or performance equivalent.
 - Performance equivalent refers to both the energy and water criteria in the ENERGY STAR product specifications, available on the ENERGY STAR website.
 - Products must meet the standards of the current version of ENERGY STAR as of the date of their purchase.

Referenced Standards

Refer to the LEED v4 reference guide, with the following modification:

Delete "Consortium for Energy Efficiency" from the list of referenced standards.

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Selecting efficient fixtures, fittings and appliances in the design phase helps ensure that projects will use less water throughout the building life cycle, which may help improve a project's water performance score.

WE Prerequisite: Building-Level Water Metering

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To conserve low cost potable water resources and support water management and identify opportunities for additional water savings by tracking water consumption.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Install permanent water meters that measure the total potable water use for the building and associated grounds. Meter data must be compiled into monthly and annual summaries; meter readings can be manual or automated.

Commit to sharing with USGBC the resulting whole-project water usage data for a five-year period beginning on the date the project accepts LEED certification or typical occupancy, whichever comes first.

This commitment must carry forward for five years or until the building changes ownership or lessee.

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide, with the following additions and modifications:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Installing whole building water meters is a simple yet critical strategy for understanding total water use throughout the building life cycle. Tracking water consumption on a regular basis supports effective water management and provides performance data to help verify that building systems are operating as designed. Projects can submit data via the Arc platform to comply with the prerequisite requirement to share whole-project water usage data with USGBC and get started on the path to recertification.

WE Credit: Outdoor Water Use Reduction

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-3 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1 point)

INTENT

To reduce outdoor potable water consumption and preserve no and low-cost potable water resources.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Reduce outdoor water use through one of the following options. Nonvegetated surfaces, such as permeable or impermeable pavement, should be excluded from landscape area calculations. Athletic fields and playgrounds (if vegetated) and food gardens may be included or excluded at the project team's discretion.

Option 1. No Irrigation Required (2 points except Healthcare and CS, 1 point Healthcare, 3 points CS)

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

Option 2. Reduced Irrigation (1-2 points except Healthcare and CS, 1 point Healthcare, 3 points CS)

Reduce the project's landscape water requirement (LWR) by at least 50% from the calculated baseline for the site's peak watering month. Reductions must first be achieved through plant species selection and irrigation system efficiency as calculated in the Environmental Protection Agency (EPA) WaterSense Water Budget Tool.

Additional reductions beyond 30% may be achieved using any combination of efficiency, alternative water sources, and smart scheduling technologies.

Table 1. Points for reducing irrigation water

<i>Percentage reduction from baseline</i>	<i>Points (except Healthcare)</i>	<i>Points (Healthcare)</i>	<i>Points (CS)</i>
50%	1	1	1
75%	--	--	2
100%	2	—	3

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent**Beta Update**

Core and Shell projects can now earn up to 3 points under WE credit Outdoor Water Use Reduction. One point was re-allocated from WE credit Indoor Water Use Reduction to this credit, to better align WE points available with the typical scope of work of Core and Shell projects and to reward incremental outdoor water use savings.

Further Explanation**Connection to Ongoing Performance**

- ▶ LEED O+M WE credit Water Performance: Designing out the need for a permanent irrigation system, selecting native or adapted plants for project landscaping and/or installing efficient irrigation systems reduces irrigation water use throughout the building life cycle, which may improve a project's water performance score. The use of alternative water systems for irrigation further reduces building potable water use.

WE Credit: Indoor Water Use Reduction

This credit applies to

- ▶ BD+C: New Construction (1-6 points)
- ▶ BD+C: Core & Shell (1-4 points)
- ▶ BD+C: Schools (1-7 points)
- ▶ BD+C: Retail (1-7 points)
- ▶ BD+C: Data Centers (1-6 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-6 points)
- ▶ BD+C: Hospitality (1-6 points)
- ▶ BD+C: Healthcare (1-7 points)

INTENT

To reduce indoor potable water consumption and preserve no and low-cost potable water resources.

REQUIREMENTS

NC, CS, SCHOOLS, NC-RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, NC-HOSPITALITY, HEALTHCARE

Further reduce fixture and fitting water use from the calculated baseline in WE Prerequisite Indoor Water Use Reduction. Additional potable water savings can be earned above the prerequisite level using alternative water sources. Include fixtures and fittings necessary to meet the needs of the occupants. Some of these fittings and fixtures may be outside the tenant space (for Commercial Interiors) or project boundary (for New Construction). Points are awarded according to Table 1.

Table 1. Points for reducing water use

Percentage Reduction	Points (BD+C)	Points (CS)	Points (Schools, Retail, Hospitality, Healthcare)
25%	1	1	1
30%	2	2	2
35%	3	3	3
40%	4	4	4
45%	5	--	5
50%	6	--	--

Schools, Retail, Hospitality, and Healthcare only

Meet the percentage reduction requirements above.

AND

Appliance and Process Water. Install equipment within the project scope that meets the minimum requirements in Table 2, 3, 4, or 5. One point is awarded for meeting all applicable requirements in any one table. All applicable equipment listed in each table must meet the standard.

Schools, Retail, and Healthcare projects can earn a second point for meeting the requirements of two tables.

Table 2. Compliant commercial washing machines

To use Table 2, the project must process at least 120,000 lbs. (57 606 kg) of laundry per year.

<i>Washing machine</i>	<i>Requirement (IP units)</i>	<i>Requirement (SI units)</i>
On-premise, minimum capacity 2,400 lbs. (1 088 kg) per 8-hour shift	Maximum 1.8 gals per pound *	Maximum 7 liters per 0.45 kilograms *

* Based on equal quantities of heavy, medium, and light soil laundry.

Table 3. Standards for commercial kitchen equipment

To use Table 3, the project must serve at least 100 meals per day of operation. All process and appliance equipment listed in the category of kitchen equipment and present on the project must comply with the standards.

<i>Kitchen equipment</i>		<i>Requirement (IP units)</i>	<i>Requirement (SI units)</i>
Dishwasher	Undercounter	ENERGY STAR	ENERGY STAR or performance equivalent
	Stationary, single tank, door	ENERGY STAR	ENERGY STAR or performance equivalent
	Single tank, conveyor	ENERGY STAR	ENERGY STAR or performance equivalent
	Multiple tank, conveyor	ENERGY STAR	ENERGY STAR or performance equivalent
	Flight machine	ENERGY STAR	ENERGY STAR or performance equivalent
Food steamer	Batch (no drain connection)	≤ 2 gal/hour/pan including condensate cooling water	≤ 7.5 liters/hour/pan including condensate cooling water
	Cook-to-order (with drain connection)	≤ 5 gal/hour/pan including condensate cooling water	≤ 19 liters/hour/pan including condensate cooling water
Combination oven,	Countertop or stand	≤ 1.5 gal/hour/pan including condensate cooling water	≤ 5.7 liters/hour/pan including condensate cooling water
	Roll-in	≤ 1.5 gal/hour/pan including condensate cooling water	≤ 5.7 liters/hour/pan including condensate cooling water
Food waste disposer	Disposer	3-8 gpm, full load condition, 10 minute automatic shutoff; or 1 gpm, no-load condition	11-30 lpm, full load condition, 10-min automatic shutoff; or 3.8 lpm, no-load condition
	Scrap collector	Maximum 2 gpm makeup water	Maximum 7.6 lpm makeup water
	Pulper	Maximum 2 gpm makeup water	Maximum 7.6 lpm makeup water

	Strainer basket	No additional water usage	No additional water usage
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gpm = gallons per minute

gph = gallons per hour

lpm = liters per minute

lph = liters per hour

Table 4. Compliant laboratory and medical equipment

To use Table 4, the project must be a medical or laboratory facility.

<i>Lab equipment</i>	<i>Requirement (IP units)</i>	<i>Requirement (SI units)</i>
Reverse-osmosis water purifier	75% recovery	75% recovery
Steam sterilizer	For 60-inch sterilizer, 6.3 gal/U.S. tray For 48-inch sterilizer, 7.5 gal/U.S. tray	For 1520-mm sterilizer, 28.5 liters/DIN tray For 1220-mm sterilizer, 28.35 liters/DIN tray
Sterile process washer	0.35 gal/U.S. tray	1.3 liters/DIN tray
X-ray processor, 150 mm or more in any dimension	Film processor water recycling unit	
Digital imager, all sizes	No water use	

Table 5. Compliant municipal steam systems

To use Table 5, the project must be connected to a municipal or district steam system that does not allow the return of steam condensate.

<i>Steam system</i>	<i>Standard</i>
Steam condensate disposal	Cool municipally supplied steam condensate (no return) to drainage system with heat recovery system or reclaimed water
OR	
Reclaim and use steam condensate	100% recovery and reuse

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Core and Shell (CS) projects now earn up to 4 points under WE credit Indoor Water Use Reduction. CS projects can now earn one additional point under WE credit Outdoor Water Use Reduction and one additional point under WE credit Optimize Process Water Use; this re-allocation intends to better align WE points available with the typical scope of work of CS projects.

Further Explanation

Occupant Types

Refer to the LEED v4 reference guide, with the following addition for projects pursuing a usage-based calculation:

If the project includes separate gender neutral and/or ADA restrooms without urinals, assume that 5% of male occupants and 5% of female occupants use these restrooms. Enter 95% into the percent of males expected to use restrooms with urinals in the Indoor Water Use Reduction Calculator. Alternately, estimate this percentage based on the project's restroom layout and anticipated usage patterns or weighted fixture counts.

Campus or Municipal Alternative Water Sources

Projects may count future infrastructure for reclaimed water systems in Water Efficiency credit calculations if:

- ▶ the future systems will be functional within 2 years of project occupancy,
- ▶ the project provides documentation, such as a signed contract, confirming that the reclaimed water will be provided to the project within a 2-year period from project occupancy, and
- ▶ the project provides confirmation that the reclaimed water supplier has agreed to supply the volume of treated wastewater or seawater claimed by the project.

International Tips

- ▶ The EU A label for residential appliances is an acceptable alternative to ENERGY STAR. The EU Ecodesign and Labelling framework establishes minimum performance standards for the energy and environmental performance of appliances and products.

Referenced Standards

Refer to the LEED v4 reference guide, with the following modification:

Delete "Consortium for Energy Efficiency" from the list of referenced standards.

Exemplary Performance

Achieve 55% savings.

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Selecting efficient fixtures, fittings and appliances in the design phase helps ensure that projects will use less water throughout the building life cycle, which may help improve a project's water performance score. The use of alternative water sources for appropriate end uses can further reduce demand for potable water and strain on the local utility.

WE Credit: Optimize Process Water Use

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-3 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To conserve low cost potable water resources used for mechanical processes while controlling, corrosion and scale in the condenser water system.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE.

Option 1. Cooling Tower and Evaporative Condenser Cycles of Concentration (1-2 points except CS, 1-3 points CS)

For cooling towers and evaporative condensers, conduct a one-time potable water analysis, measuring at least the five control parameters listed in Table 1.

Table 1. Maximum concentrations for parameters in condenser water

<i>Parameter</i>	<i>Maximum level</i>
Ca (as CaCO ₃)	600 ppm
Total alkalinity	500 ppm
SiO ₂	150 ppm
Cl ⁻	300 ppm
Conductivity	3300 μS/cm

ppm = parts per million

μS/cm = micro siemens per centimeter

Calculate the maximum number of cooling tower cycles by dividing the maximum allowed concentration level of each parameter by the actual concentration level of each parameter found in the potable makeup water analysis. Limit cooling tower cycles to avoid exceeding maximum values for any of these parameters.

The materials of construction for the water system that come in contact with the cooling tower water shall be of the type that can operate and be maintained within the limits established in Table 1.

Table 2. Points for cooling tower cycles

<i>Cooling tower cycles</i>	<i>Points (all except CS)</i>	<i>Points (CS)</i>
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Maximum number of cycles achieved without exceeding any maximum concentration levels or affecting operation of condenser water system	1	1
Meet the maximum calculated number of cycles to earn 1 point, and increase the number of cycles by a minimum of 25% by increasing the level of treatment and/or maintenance in condenser or make-up water systems OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 20% recycled nonpotable water	2	2
Meet the maximum calculated number of cycles to earn 1 point, and increase the number of cycles by a minimum of 30% by increasing the level of treatment and/or maintenance in condenser or make-up water systems OR Meet the maximum calculated number of cycles to earn 1 point and use a minimum 30% recycled nonpotable water	--	3

Minimum percentage recycled nonpotable water used in cooling tower makeup should be based on water use during the month with the highest demand for make-up water.

Projects may consider using water reclaimed from another process as a source of makeup water for evaporative cooling equipment as long as the resultant circulating water chemistry conforms to the parameters established in Table 1.

Projects whose cooling is provided by district cooling systems are eligible to achieve Option 1 if the district cooling system complies with the above requirements.

OR

Option 2. Optimize Water Use for Cooling (1-2 points except CS, 1-3 points CS)

To be eligible for Option 2, the baseline system designated for the building using ASHRAE 90.1-2016 Appendix G Table G3.1.1 must include a cooling tower (systems 7, 8, 11, 12, and 13)

Achieve increasing levels of cooling tower water efficiency beyond a water-cooled chiller system with axial variable-speed fan cooling towers having a maximum drift of 0.002% of recirculated water volume and three cooling tower cycles. Points are awarded according to Table 3.

Table 3. Points for reducing annual water use compared to Water-Cooled Chiller System

Percentage Reduction	Points (BD+C)	Points (CS)
25%	1	1
50%	2	2
100%	-	3

Projects whose cooling is provided by district cooling systems are eligible to achieve Option 2 if the district cooling system complies with the above requirements.

AND/OR

Option 3. Process Water Use (1-2 points except CS, 1-3 points CS)

Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, or using minimum 30% recycled alternative water to meet process water demand for 2 points. Ensure that recycled alternative water is of sufficient quality for its intended end use.

Minimum percentage of recycled alternative water used should be based on water use during the month with the highest water demand.

Process water uses eligible for achievement of Option 3 must represent at least 10% of total building regulated water use and may not include water used for cooling. Eligible subsystems may include:

- ▶ Boilers
- ▶ Humidification systems
- ▶ Other subsystems using process water

Projects served by district systems are eligible to achieve Option 3 if the district system complies with minimum thresholds for recycled alternative water use.

Core and Shell projects:

Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, using minimum 30% recycled alternative water to meet process water demand for 2 points, or using minimum 40% recycled alternative water to meet process water demand for 3 points. Ensure that recycled alternative water is of sufficient quality for its intended end use.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Evaporative cooling is the most energy efficient method to reject heat from a process (1 lb. of water removes almost 1,000 BTU's (970 BTU/lb.) with the added benefit of reducing greenhouse gas emissions. Cooling towers and evaporative condensers utilize evaporative cooling to operate with maximum energy efficiency; to preserve this energy savings, it is important to conserve the water these system use, especially in water stressed regions. In some climate zones, heat rejection system designs can help to decrease the water intensity of the building cooling process using a combination of wet, dry, or adiabatic cooling. In some cases, 100% dry operation is possible.

For wet cooling systems, this credit addresses water conservation strategies while still saving energy. This strategy includes maximizing cycles of concentration in cooling towers and evaporative condensers and make-up water reduction strategies including the use of reclaimed water and hybrid cooling. The use of alternative recycled water to meet process water demands can further reduce building potable water use.

Updates to this credit recognize regional variations in the quality of local water supply and recognize additional strategies for reducing potable water use for diverse project types.

Core and Shell projects can now earn up to 3 points under WE credit Optimize Process Water Use. One point was re-allocated from WE credit Indoor Water Use Reduction to this credit, to better align WE points available with the typical scope of work of Core and Shell projects and to reward incremental process water savings.

Step-by-Step Guidance

Option 1. Cooling Tower and Evaporative Condenser Cycles of Concentration

Refer to the v4 BD+C reference guide with the following addition:

Step 6. Address construction materials for cooling system

Share the results of the potable water analysis with the cooling system manufacturer and ensure that the materials of construction for all components of the water-cooled system that come in contact with the cooling loop can operate and be maintained within the limits established in Table 1. The evaporative cooled system equipment must be designed to accommodate the targeted cycles of concentration calculated for the cooling tower or evaporative condenser.

Option 2. Optimize Water Use for Cooling

Option 2 is limited to projects that use water-cooled systems in the ASHRAE 90.1-2016 baseline design. ASHRAE 90.1 Appendix G identifies baseline system types that are standard practice for similar, newly constructed buildings.

Step 1. Determine baseline system for building cooling.

Confirm that at least one of the baseline system(s) designated for the building using ASHRAE 90.1-2016 Appendix G Table G3.1.1 includes a cooling tower or evaporative condenser (systems 7, 8, 11, 12, and 13).

If the above conditions are met and the project team eliminates the need for cooling towers or evaporative condensers in the proposed building design, then the project team may document a 100% reduction in water use for cooling.

Step 2. Develop building cooling systems design.

Consider viable on-site alternatives to 100% evaporative cooled water-cooled systems for the building and the impacts of the cooling systems design on building energy use. Hybrid water-cooled system options may be used to reduce water usage while saving energy.

Step 3. Document compliance.

If the project includes systems that use the latent heat of evaporative cooling, calculate annual water savings as compared to a 100% water-cooled chiller system, based on proposed cooling loads for spaces served by systems 7, 8, 11, 12, or 13 in the baseline. The system performance for the reference shall be calculated based on a 100% water-cooled chiller system with axial variable-speed fan cooling towers having a maximum drift of 0.002% of recirculated water volume and three cooling tower cycles.

Some energy simulation software has the capability of simulating cooling tower water usage within the software. To perform the calculations within the energy model, the proposed case model shall be modeled consistent with the project design to demonstrate the total water usage associated with evaporative cooling in the proposed design.

To determine the reference case performance, the proposed case model shall be updated with a water-cooled chilled water system serving all spaces served by systems 7, 8, 11, 12, or 13 in the ASHRAE 90.1 Baseline model. All proposed parameters (such as fan power, fan control, economizer control, demand control ventilation, energy recovery, dedicated outside air supply, quantity of systems, etc.) shall remain identical, except that the cooling type shall be revised to water-cooled chillers matching those stipulated in ASHRAE 90.1 Appendix G.

If performing the cooling tower water usage calculations outside the energy model, cooling tower and evaporative condenser manufacturers have water usage programs based on design conditions, building load and water chemistry. To calculate the water usage with the manufacturer's data, the design cooling loads from the proposed model for spaces served by systems 7, 8, 11, 12, or 13 in the baseline shall be

exported from the energy model, and supplemental calculations shall be documented based on the proposed building cooling load and water use data sheets from equipment suppliers. Any assumptions used for the calculations shall be submitted with the calculations.

The manufacturer's program used for calculating water usage must incorporate the hourly load profile from the energy model or be based on climate-specific load profiles for the project's building type. If hourly load profiles and hourly weather profiles are not used, the calculations shall include "bins" that account for variations in the load profile and water usage dependent on outdoor air drybulb temperature, outdoor air wetbulb temperature, and building cooling load. The load profile must be identical in the proposed and reference case.

Option 3. Process Water Use

Step 1. Identify water subsystems.

Eligible subsystems may include boilers, humidification systems, or other subsystems using process water that represent at least 10% of total regulated building water use. Water used for cooling is not eligible for Option 3.

- ▶ Consider minimum required quality of make-up water for water subsystems.

Step 2. Identify sources of recycled alternative water.

Identify recycled alternative water sources that could be used to meet process water subsystem demand.

Alternative water sources include municipally supplied reclaimed wastewater ("purple pipe" water), graywater, rainwater, stormwater, treated seawater, water recovered from condensate, foundation dewatering water, treated blowdown from process water, reverse osmosis reject water, and other recycled water sources. Well water, groundwater, and naturally occurring surface bodies of water (such as streams, lakes, or rivers) do not contribute to recycled alternative water sources.

Step 3. Calculate subsystem water use.

Calculate the water use of subsystems using manufacturer information and anticipated operating conditions.

- ▶ To be eligible for Option 3, the subsystem must represent at least 10% of total building regulated water use annually. Total building water includes all indoor water consumption associated with the project in the design case and excludes outdoor landscape water use.
- ▶ Calculate subsystem water use during each month; identify the month with the highest water demand.

Step 4. Calculate quantity of recycled alternative water source.

Calculate the quantity of alternative water sources available for reuse each month. Address the cistern storage capacity for on-site alternative water systems. For municipally supplied alternative water systems, demonstrate that the municipality has approved to supply the volume of recycled water required by the project.

Step 5. Select alternative water source to meet subsystem demand.

Confirm that alternative water source is of sufficient quantity to meet at least 20% of water subsystem demand during the month with the highest water demand. If the alternative water is used for multiple applications—for example, boilers, flush fixtures, and landscape irrigation—a sufficient quantity must be available to meet the demands of all uses. Teams cannot apply the same alternative water to multiple credits unless the recycled alternative water source has sufficient volume to cover the demand of all the uses (e.g., boilers, irrigation plus toilet-flushing demand).

Confirm that alternative water source is of sufficient quality to meet intended use or treat alternative water source.

- ▶ Minimum requirements for make-up water quality vary by subsystem.
- ▶ When selecting alternative sources of water, target the uses that require the least treatment first.
- ▶ As needed, treat alternative water sources to be of sufficient quality for intended end use.

Project Type Variations

District Energy Systems

Option 2. Optimize Water Use for Cooling

For projects connected to a district cooling system, the project shall demonstrate that any on-site cooling equipment meets the credit requirements and that upstream systems meet the credit requirements.

To pursue this option, the baseline system designated for the building in accordance with ASHRAE 90.1-2016 Appendix G Table G3.1.1 (prior to modeling purchased energy) must include a cooling tower (systems 7 &, 8, 11, 12, and 13).

For upstream equipment from the District Energy System, provide a narrative describing how the system is operated and specifically describing the district energy cooling equipment, including capacities, equipment type, etc.

If cooling towers, evaporative condensers, evaporative cooling systems, and other evaporative cooling systems are not used in the district cooling system serving the project, or are used in conjunction with compressor-based cooling equipment to supply no more than 10% of the total cooling energy generated by the District Energy System, then the project automatically qualifies for 2 points under the credit (Core and Shell projects automatically qualify for 3 points) and does not need to provide calculations. Provide a narrative description of how the district cooling system is operated and confirm that there is no evaporative cooling equipment in the district energy system.

If the district cooling system does use evaporative cooling, the project team must provide calculations demonstrating achievement of the Option 2 requirements based on total district cooling capacity and cooling load profile for the entire district system (building cooling loads shall not be used to document performance).

The program or supplemental calculations used for calculating water usage must either incorporate the hourly load profile from the District Energy System, or include “bins” that account for variations in the District Energy Plant load profile and water usage dependent on outdoor air drybulb temperature, outdoor air wetbulb temperature, and District Energy thermal cooling load. The load profile must be identical in the proposed and reference case.

Option 3. Process Water Use

For projects where heating is provided by a district heating system, the project team must pro-rate the district heating system’s total water usage to the building to determine whether it represents at least 10% of total building regulated water use.

For example, if the district heating system consumes 500,000 gallons of water annually and the LEED project uses 1% of the district heating system annual heat generation, the total prorated water consumption from the district heating system is 500,000 gallons x 1% = 5,000 gallons. This must represent at least 10% of the total building water consumption, meaning that the total building consumption must be less than or equal to 50,000 gallons.

Further Explanation

Required Documentation

Documentation	Option 1		Option 2	Option 3
	1 point	2 points		
Potable water analysis results	x	X		
Potable water analysis narrative	x	X		
Cycles of concentration calculations	x	X		
Recycled Nonpotable water calculations		X		
Water treatment calculations		X		
Nonpotable water analysis (if using 100% nonpotable water)		X		
Documentation showing that project Baseline system is designated as systems 7, 8, 11, 12, or 13 under ASHRAE 90.1-2016 Appendix G Table G3.1.3			x	
Calculations demonstrating percent reduction in cooling tower water usage for systems where the proposed design uses the latent heat of evaporative cooling of water			x	
Site or mechanical systems plan, energy model or other showing project design			x	x
Water subsystem monthly demand calculations				x
Recycled alternative water source quantity calculations and plumbing drawings/schematics of the alternative water system. For municipally supplied alternative water, provide documentation that the municipality has agreed to supply the volume of recycled alternative water claimed by the project				x
Manufacturer information for water subsystem				x

Exemplary Performance

- ▶ Option 2. Demonstrate a 100% reduction in annual water use compared to a water-cooled chiller system.
- ▶ Option 3. Use minimum 40% recycled alternative water to meet process water demand (all projects except Core and Shell). Core and Shell: use minimum 50% recycled alternative water to meet process water demand.

Referenced Standards

- ▶ ASHRAE Standard 90.1-2016

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Designing building cooling systems and other water subsystems to minimize potable water and reuse alternative water sources can significantly reduce the project's water footprint over the building life cycle, which may help improve a project's water performance score. Additionally, treating and maintaining the quality of makeup water used to meet process water demands can preserve the performance and efficiency of water using subsystems, reducing the frequency of replacement and repairs.

WE Credit: Water Metering

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To conserve low cost potable water resources and support water management and identify opportunities for additional water savings by tracking water consumption.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Install permanent water meters for two or more of the following water subsystems, as applicable to the project:

- ▶ *Irrigation.* Meter water systems serving at least 80% of the irrigated landscaped area. Calculate the percentage of irrigated landscape area served as the total metered irrigated landscape area divided by the total irrigated landscape area. Landscape areas fully covered with xeriscaping or native vegetation that requires no routine irrigation may be excluded from the calculation.
- ▶ *Indoor plumbing fixtures and fittings.* Meter water systems serving at least 80% of the indoor fixtures and fitting described in WE Prerequisite Indoor Water Use Reduction, either directly or by deducting all other measured water use from the measured total water consumption of the building and grounds.
- ▶ *Domestic hot water.* Meter water use of at least 80% of the installed domestic hot water heating capacity (including both tanks and on-demand heaters).
- ▶ *Boiler with aggregate projected annual water use of 100,000 gallons (378 500 liters) or more, or boiler of more than 500,000 BtuH (150 kW).* A single makeup meter may record flows for multiple boilers.
- ▶ *Reclaimed water.* Meter reclaimed water, regardless of rate. A reclaimed water system with a makeup water connection must also be metered so that the true reclaimed water component can be determined.
- ▶ *Other process water.* Meter at least 80% of expected daily water consumption for process end uses, such as humidification systems, dishwashers, clothes washers, pools, and other subsystems using process water.

Healthcare Projects only

In addition to the requirements above, install water meters in any five of the following:

- ▶ purified water systems (reverse-osmosis, de-ionized);
- ▶ filter backwash water;
- ▶ water use in dietary department;
- ▶ water use in laundry;
- ▶ water use in laboratory;
- ▶ water use in central sterile and processing department;

- ▶ water use in physiotherapy and hydrotherapy and treatment areas;
- ▶ water use in surgical suite;
- ▶ closed-looped hydronic system makeup water; and
- ▶ cold-water makeup for domestic hot water systems.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following additions and modifications:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Submetering water subsystems helps facility managers track changes in water usage over time and provides the data necessary to identify opportunities for water savings by end use, which may help improve a project's water performance score. Submetering is an important component of a successful water management program; metered data enables monitoring of consumption and costs as well as progress reporting throughout the building life cycle.